# JEE Mains 2019 Chapter wise Question Bank

## Kinetic Theory of Gases - Questions

**Q1**

A mixture of 2 moles of helium gas (atomic mass = 4u), and 1 mole of argon gas (atomic mass = 40u) is kept at 300 K in a container. The ratio of their rms speeds $\left[\frac{V_{rms}(\text{helium})}{V_{rms}(\text{argon})}\right]$ is close to :

(1) 3.16 (2) 0.32
(3) 0.45 (4) 2.24

**9 Jan Morning**

**Q2**

A 15 g mass of nitrogen gas is enclosed in a vessel at a temperature 27°C. Amount of heat transferred to the gas, so that rms velocity of molecules is doubled, is about: [Take R = 8.3 J/K mole]

(1) 0.9 kJ (2) 6 kJ
(3) 10 kJ (4) 14 kJ

**9 Jan Evening**

**Q3**

Two kg of a monoatomic gas is at a pressure of $4 \times 10^4$ N/m$^2$. The density of the gas is 8 kg/m$^3$. What is the order of energy of the gas due to its thermal motion?

(1) $10^3$ J (2) $10^5$ J
(3) $10^4$ J (4) $10^6$ J

**10 Jan Evening**

**Q4**

A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Considering only translational and rotational modes, the total internal energy of the system is :

(1) 15 RT (2) 12 RT
(3) 4 RT (4) 20 RT

**11 Jan Morning**

**Q5**

An ideal gas occupies a volume of $2\ m^3$ at a pressure of $3\times10^6$ Pa. The energy of the gas:

(1) $9\times10^6$ J (2) $6\times10^4$ J
(3) $10^8$ J (4) $3\times10^2$ J

**12 Jan Morning**

**Q6**

An ideal gas is enclosed in a cylinder at pressure of 2 atm and temperature, 300 K. The mean time between two successive collisions is $6 \times 10^{-8}$ s. If the pressure is doubled and temperature is increased to 500 K, the mean time between two successive collisions wiil be close to :

(1) $2 \times 10^{-7}$ s (2) $4 \times 10^{-8}$ s
(3) $0.5 \times 10^{-8}$ s (4) $3 \times 10^{-6}$ s

**12 Jan Evening**

**Q7**

If $10^{22}$ gas molecules each of mass $10^{-26}$ kg collide with a surface (perpendicular to it) elastically per second over an area 1 m$^2$ with a speed $10^4$ m/s, the pressure exerted by the gas molecules will be of the order of :

(1) $10^4$ N/m$^2$ (2) $10^8$ N/m$^2$
(3) $10^3$ N/m$^2$ (4) $10^{16}$ N/m$^2$

**8 April Morning**

**Q8**

The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to :

[Boltzmann Constant $k_B = 1.38 \times 10^{-23}$ J/K

Avogadro Number $N_A = 6.02 \times 10^{26}$ /kg

Radius of Earth : $6.4 \times 10^6$ m

Gravitational acceleration on Earth = 10 ms$^{-2}$]

(1) 800 K (2) $3 \times 10^5$ K
(3) $10^4$ K (4) 650 K

**8 April Evening**

**Q9**

For a given gas at 1 atm pressure, rms speed of the molecules is 200 m/s at 127°C. At 2 atm pressure and at 227°C, the rms speed of the molecules will be:

(1) 100 m/s (2) $80\sqrt{5}$ m/s
(3) $100\sqrt{5}$ m/s (4) 80 m/s

**9 April Morning**

**Q10**

An HCl molecule has rotational, translational and vibrational motions. If the rms velocity of HCl molecules in its gaseous phase is $\bar{v}$, m is its mass and $k_B$ is Boltzmann constant, then its temperature will be:

(1) $\frac{m\bar{v}^2}{6k_B}$ (2) $\frac{m\bar{v}^2}{3k_B}$ (3) $\frac{m\bar{v}^2}{7k_B}$ (4) $\frac{m\bar{v}^2}{5k_B}$

**9 April Morning**

**Q11**

The specific heats, $C_p$ and $C_v$ of a gas of diatomic molecules, A, are given (in units of J mol$^{-1}$ k$^{-1}$) by 29 and 22, respectively. Another gas of diatomic molecules, B, has the corresponding values 30 and 21. If they are treated as ideal gases, then:
(1) A is rigid but B has a vibrational mode.
(2) A has a vibrational mode but B has none.
(3) A has one vibrational mode and B has two.
(4) Both A and B have a vibrational mode each.

**9 April Evening**

**Q11**

A $25\times10^{-3}$ m$^3$ volume cylinder is filled with 1 mol of $O_2$ gas at room temperature (300 K). The molecular diameter of $O_2$, and its root mean square speed, are found to be 0.3 nm and 200 m/s, respectively. What is the average collision rate (per second) for an $O_2$ molecule?
(1) $\sim10^{12}$ (2) $\sim10^{11}$ (3) $\sim10^{10}$ (4) $\sim10^{13}$

**10 April Morning**

**Q12**

Two moles of helium gas is mixed with three moles of hydrogen molecules (taken to be rigid). What is the molar specific heat of mixture at constant volume? (R = 8.3 J/mol K)
(1) 19.7 J/mol L (2) 15.7 J/mol K
(3) 17.4 J/mol K (4) 21.6 J/mol K

**12 April Morning**

## JEE Mains 2019 Chapter wise Question Bank

### Kinetic Theory of Gases - Answers

**Q1**

**(1)** Using $\frac{V_{1rms}}{V_{2rms}} = \sqrt{\frac{M_2}{M_1}}$

$$\frac{V_{rms}(He)}{V_{rms}(Ar)} = \sqrt{\frac{M_{Ar}}{M_{He}}} = \sqrt{\frac{40}{4}} = 3.16$$

**9 Jan Morning**

**Q2**

**(3)** Heat transferred,

$Q = nC_V \Delta T$ as gas in closed vessel

To double the rms speed, temperature should be 4 times i.e.,

$T' = 4T$ as $v_{rms} = \sqrt{3RT/M}$

$$\therefore Q = \frac{15}{28} \times \frac{5 \times R}{2} \times (4T - T)$$

$$\left[\therefore \frac{CP}{CV} = \gamma_{diatomic} = \frac{7}{5} \& C_p - C_v = R\right]$$

or, $Q = 10000\,J = 10\,kJ$

**9 Jan Evening**

**Q3**

**(3)** Thermal energy of N molecule

$$= N\left(\frac{3}{2}kT\right)$$

$$= \frac{N}{N_A}\frac{3}{2}RT = \frac{3}{2}(nRT) = \frac{3}{2}PV$$

$$= \frac{3}{2}P\left(\frac{m}{\rho}\right) = \frac{3}{2}P\left(\frac{2}{8}\right)$$

$$= \frac{3}{2} \times 4 \times 10^4 \times \frac{2}{8} = 1.5 \times 10^4$$

therefore, order $= 10^4$

**10 Jan Evening**

**Q4**

**(1)** $U = \frac{f_1}{2}n_1RT + \frac{f_2}{2}n_2RT$

Considering translational and rotational modes, degrees of freedom $f_1 = 5$ and $f_2 = 3$

$\therefore u = \frac{5}{2}(3RT) + \frac{3}{2} \times 5RT$

$U = 15RT$

**11 Jan Morning**

**Q5**

**(1)** Energy of the gas, E

$$= \frac{f}{2}nRT = \frac{f}{2}PV$$

$$= \frac{f}{2}(3 \times 10^6)(2) = f \times 3 \times 10^6$$

Considering gas is monoatomic i.e., $f = 3$

Energy, $E = 9 \times 10^6$ J

**12 Jan Morning**

**Q6**

**(2)** Using, $\tau = \frac{1}{2n\pi d^2 V_{avg}}$

$$\therefore t \propto \frac{\sqrt{T}}{P}\left[\because n = \frac{\text{no.of molecules}}{\text{Volume}}\right]$$

$$\text{or, } \frac{t_1}{6\times10^{-8}} = \frac{\sqrt{500}}{2P}\times\frac{P}{\sqrt{300}} \approx 4\times10^{-8}$$

**12 Jan Evening**

**Q7**

**(Bouns)**

Rate of change of momentum during collision

$$= \frac{mv-(-mv)}{\Delta t} = \frac{2mv}{\Delta t}N$$

so pressure

$$P = \frac{N\times(2mv)}{\Delta t\times A}$$

$$= \frac{10^{22}\times2\times10^{-26}\times10^4}{1\times1} = 2\,N/m^2$$

**8 April Morning**

**Q8**

**(3)** $v_{rms} = v_e$

$$\sqrt{\frac{3RT}{M}} = 11.2\times10^3$$

$$\text{or } \sqrt{\frac{3kT}{m}} = 11.2\times10^3$$

$$\text{or } \sqrt{\frac{3\times1.38\times10^{-23}T}{2\times10^{-3}}} = 11.2\times10^3 \quad \therefore v = 10^4\,\text{K}$$

**8 April Evening**

**Q9**

**(3)** $V_{rms} = \sqrt{\frac{3RT}{M}}$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{(273+127)}{(273+227)}}$$

$$= \sqrt{\frac{400}{500}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$$

$$\therefore v_2 = \frac{\sqrt{5}}{2}v_1 = \frac{\sqrt{5}}{2}\times200 = 100\sqrt{5}\text{ m/s.}$$

**9 April Morning**

**Q10**

**(1)** In this case the total degree of freedom is 6.

According to law of equipartition of energy,

$$\frac{1}{2}m\bar{v}^2 = 6\left(\frac{1}{2}k_BT\right)$$

$$\therefore \frac{1}{2}m\bar{v}^2 = 3k_BT$$

$$\text{or, } T = \frac{m\bar{v}^2}{6k_B}$$

**9 April Morning**

**Q11**

**(2)** $\gamma_A = \frac{C_P}{C_v} = \frac{29}{22} = 1.32 < 1.4$ (diatomic)

and $\gamma_B = \frac{30}{21} = \frac{10}{7} = 1.43 > 1.4$

Gas A has more than 5-degrees of freedom.

**9 April Evening**

**Q11**

**(3)** $V = 25\times10^{-3}\text{ m}^3$, N = 1 mole of $O_2$

T = 300 K

$V_{rms} = 200$ m/s

$$\therefore \lambda = \frac{1}{\sqrt{2}N\pi r^2}$$

$$\text{Average time } \frac{1}{\tau} = \frac{\langle V\rangle}{\lambda} = 200.N\pi r^2.\sqrt{2}$$

$$= \frac{\sqrt{2}\times200\times6.023\times10^{23}}{25\times10^{-3}}.\pi\times10^{-18}\times0.09$$

The closest value in the given option is $= 10^{10}$

**10 April Morning**

**Q12**

**(3)** $[C_v]_{\min} = \frac{n_1[C_{v_1}] + n_2[C_{v_2}]}{n_1 + n_2}$

$= \left[\frac{2 \times \frac{3R}{2} + 3 \times \frac{5R}{2}}{2+3}\right]$

$= 2.1\ \text{R}$

$= 2.1 \times 8.3$

$= 17.4$ J/mol–k

**12 April Morning**